

Please try to attempt both questions below but submit only **ONE** as your assignment. If a question is not clear, make and state appropriate assumptions and solve it. You're allowed to consult your peers or the internet. However, I would encourage you to attempt to solve the questions first before consulting. There *may* be a quiz from this later.

Question 1 (Marks:10): Imagine a situation where Agent A is responsible for securing a sensitive network, while an adversarial Agent B wants to hack it. Assume both agents are perfectly optimal and rational and are aware of what possible choices are available to each of them at every step. Answer the following questions:

- a) **(Marks: 3)** Construct a Game Tree from Agent A's perspective. The game proceeds for four moves (also known as plies).
- Agent A chooses one of two possible defense measures.
 - Agent B chooses one of two possible attack measures.
 - Agent A chooses one of two possible defense measures.
 - Agent B chooses one of two possible attack measures.

At the end Agent A can achieve a Security score from the following (ordered left to right for the leaves):

3 8 1 -5 10 -10 7 6 2 1 10 9 -8 7 5 6

Please use conventions that we covered in class to represent max and min nodes as well as the leaves.

- b) **(Marks: 3)** Use the MiniMax Algorithm to calculate the value of the root as well as all the intermediate nodes. You can show the steps of the algorithm by crossing out older values of the nodes and writing new values as you move forward.
- c) **(Marks: 1)** On the tree mark the optimum path that the defense and attack will take.
- d) **(Marks: 3)** Use alpha-beta pruning to try and prune branches from this tree. You can show the steps of the algorithm by crossing out older values of the nodes and writing new values as you move forward. Please redraw the tree and either cross out unopened branches or don't show them.

Question 2 (Marks:10): Imagine a situation involving two trading agents. Agent A aims to trade in such a way as to maximize their profits, while Agent B is a competitor that wants to manipulate the markets so as to minimize Agent A's profits. The two actions available to Agent A are **Hold** or **Sell**. The two actions available to Agent B are **Aggressively Buy** or **Aggressively Sell**. Assume both agents are perfectly optimal and rational and are aware of what possible choices are available to each of them at every step. Answer the following questions:

a) (Marks: 3) Construct a Game Tree from **Agent B's perspective**. The game proceeds for four moves (also known as plies).

- Agent B chooses to buy or sell.
- Agent A chooses to hold or sell.
- Agent B chooses ...
- Agent A chooses ...

At the end Agent A can generate a profit or a loss of following values (ordered left to right for the leaves):

65, 20, 0, -50, 34, -100, 78, 33, 21, 11, 110, 95, -89, 27, 15, 16

Please use conventions that we covered in class to represent max and min nodes as well as the leaves.

- b) (Marks: 3)** Use the MiniMax Algorithm to calculate the value of the root as well as all the intermediate nodes. You can show the steps of the algorithm by crossing out older values of the nodes and writing new values as you move forward.
- c) (Marks: 1)** On the tree mark the optimum path that the two agents will take.
- d) (Marks: 3)** Use alpha-beta pruning to try and prune branches from this tree. You can show the steps of the algorithm by crossing out older values of the nodes and writing new values as you move forward. Please redraw the tree and either cross out unopened branches or don't show them.